

线性规划

姓名：李盛鹏 学号：2351136 日期：2025年6月15日

问题描述

目标函数（最小化）：

$$\min z = x_1 + x_2 + x_3$$

约束条件：

$$\begin{cases} 2x_1 + 7.5x_2 + 3x_3 \geq 10000 \\ 20x_1 + 5x_2 + 10x_3 \geq 30000 \\ x_1, x_2, x_3 \geq 0 \end{cases}$$

C++实现代码

```
#include<iostream>
#include<algorithm>
#include<cstring>
#include<cstdio>
#include<cmath>
#include <vector>
using namespace std;

const double eps = 1e-10;
enum
{
    mxn = 50, mxm = 50
}; // 适当调整大小

class Simplex
{
public:
    int n, m, t;
    double c[mxn];
    double a[mxm][mxn];
    int idx[mxn], idy[mxn];
    int st[mxn], top = 0;

    Simplex(int _m, int _n)
    {
        this->m = _m;
```

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    this->n = _n;
    for (int i = 1; i <= n; i++) idx[i] = i; // 基变量
    for (int i = 1; i <= m; i++) idy[i] = i + n; // 非基变量
}

void set_objective(double ci[mxn])
{
    for (int i = 1; i <= n; i++)
    {
        a[0][i] = -ci[i - 1]; // 第0行作为目标函数
    }
}

void set_co_matrix(double co_matrix[mxm][mxn])
{
    for (int i = 1; i <= m; i++)
    {
        for (int j = 1; j <= n; j++)
        {
            a[i][j] = co_matrix[i - 1][j - 1];
        }
    }
}

void set_bi_matrix(double b_matrix[mxm])
{
    for (int i = 1; i <= m; i++)
    {
        a[i][0] = b_matrix[i - 1]; // 第0列作为b矩阵
    }
    a[0][0] = 0;
}

int init_simplex()
{
    while (1)
    {
        int i, x = 0, y = 0;
        for (i = 1; i <= m; i++)
        {
            if (a[i][0] < -eps && ((!x) || (rand() & 1)))
            {
                x = i;
            }
        }
        if (!x) break;
    }
}

```

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    for (i = 1; i <= n; i++)
    {
        if (a[x][i] < -eps && ((!y) || (rand() & 1)))
        {
            y = i;
        }
    }

    if (!y)
    {
        printf("Infeasible\n");
        return 0;
    }
    Pivot(x, y);
}
return 1;
}

void Pivot(int x, int y)
{
    swap(idy[x], idx[y]);
    double tmp = a[x][y];
    a[x][y] = 1 / a[x][y];
    top = 0;

    for (int i = 0; i <= n; i++)
    {
        if (y != i) a[x][i] /= tmp;
    }

    for (int i = 0; i <= n; i++)
    {
        if ((y != i) && fabs(a[x][i]) > eps)
        {
            st[++top] = i;
        }
    }

    for (int i = 0; i <= m; i++)
    {
        if ((i == x) || (fabs(a[i][y]) < eps)) continue;

        for (int j = 1; j <= top; j++)
        {
            a[i][st[j]] -= a[x][st[j]] * a[i][y];

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        }
        a[i][y] = -a[i][y] / tmp;
    }
}

int run()
{
    int init = init_simplex();
    if (init == 0) return init;

    while (1)
    {
        int x = 0, y = 0;
        double mn = 1e15;

        for (int i = 1; i <= n; i++)
        {
            if (a[0][i] > eps)
            {
                y = i;
                break;
            }
        }
        if (!y) break;

        for (int i = 1; i <= m; i++)
        {
            if (a[i][y] > eps && (a[i][0] / a[i][y] < mn))
            {
                mn = a[i][0] / a[i][y];
                x = i;
            }
        }

        if (!x)
        {
            printf("Unbounded\n");
            return -1;
        }
        Pivot(x, y);
    }
    return 1;
}

pair<vector<double>, double> getans()
{

```

```

        vector<double> x;
        double z = a[0][0];

        for (int i = 1; i <= n; i++) a[0][i] = 0;
        for (int i = 1; i <= m; i++)
        {
            if (idy[i] <= n) a[0][idy[i]] = a[i][0];
        }
        for (int i = 1; i <= n; i++)
        {
            x.push_back(a[0][i]);
        }
        return make_pair(x, z);
    }
};

int main()
{
    // 问题数据
    double b_matrix[mxm] = { -10000, -30000 }; // 注意这里是负号, 因为原约束是≥
    double co_matrix[mxm][mxn] = {
        { -2, -7.5, -3 }, //  $2x_1 + 7.5x_2 + 3x_3 \geq 10000 \rightarrow -2x_1 - 7.5x_2 - 3x_3 \leq -10000$ 
        { -20, -5, -10 } //  $20x_1 + 5x_2 + 10x_3 \geq 30000 \rightarrow -20x_1 - 5x_2 - 10x_3 \leq -30000$ 
    };
    double c_matrix[mxn] = { 1, 1, 1 }; //  $\min z = x_1 + x_2 + x_3$ 

    Simplex simplex(2, 3); // 2个约束, 3个变量
    simplex.set_objective(c_matrix);
    simplex.set_co_matrix(co_matrix);
    simplex.set_bi_matrix(b_matrix);

    int result = simplex.run();

    if (result == 1)
    {
        pair<vector<double>, double> rst = simplex.getans();
        cout << "最优解: z = " << rst.second << endl;
        cout << "x1 = " << rst.first[0] << endl;
        cout << "x2 = " << rst.first[1] << endl;
        cout << "x3 = " << rst.first[2] << endl;
    }
    else if (result == 0)
    {
        cout << "问题无可行解" << endl;
    }
}

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    }  
    else  
    {  
        cout << "问题无界" << endl;  
    }  
  
    return 0;  
}
```

计算过程

2351136

李盛鹏

$$\min Z = x_1 + x_2 + x_3$$

$$\Rightarrow \max Z = -(x_1 + x_2 + x_3)$$

$$2x_1 + 7.5x_2 + 3x_3 \geq 10000$$

$$20x_1 + 5x_2 + 10x_3 \geq 30000$$

$$x_1, x_2, x_3 \geq 0$$

加入松弛变量, 则 $2x_1 + 7.5x_2 + 3x_3 - y_1 = 10000$

$$20x_1 + 5x_2 + 10x_3 - y_2 = 30000$$

$$x_1, x_2, x_3, y_1, y_2 \geq 0$$

由于找不到单位阵, 故加入人工变量

$$\text{则 } z_1 + 2x_1 + 7.5x_2 + 3x_3 - y_1 = 10000$$

$$z_2 + 20x_1 + 5x_2 + 10x_3 - y_2 = 30000$$

$$\text{需满足 } \max z' = -(z_1 + z_2) = 0$$

阶段: 代入, 只需求 $\max z' = -40000 + 22x_1 + 12.5x_2 + 13x_3 - y_1 - y_2$

	x_1	x_2	x_3	y_1	y_2
z'	-40000	22	12.5	13	-1 -1
z_1	10000	2	7.5	3	-1 0
z_2	30000	20	5	10	0 -1

由 $\min\{\frac{10000}{2}, \frac{30000}{20}\} = \frac{30000}{20}$

将 x_1, z_2 交换,

将 $x_1 = \frac{30000 - z_2 - 5x_2 - 10x_3 + y_2}{20}$ 代入

	z_2	x_2	x_3	y_1	y_2
z'	-7000	-1.1	7	2	-1 0.1
z_1	7000	-0.1	7	2	-1 0.1
x_1	1500	0.05	0.25	0.5	0 -0.05

由 $\min\{\frac{7000}{7}, \frac{1500}{0.25}\} = \frac{7000}{7}$

将 z_1, x_2 交换,

将 $x_2 = \frac{7000 - z_1 + 0.1z_2 - 2x_3 + y_1 - 0.1y_2}{7}$

代入

	z_2	x_2	x_3	y_1	y_2
z'	0	-1	-1	0	0 0
x_2	1000	$-\frac{1}{70}$	$\frac{1}{7}$	$\frac{2}{7}$	$-\frac{1}{7}$ $\frac{1}{70}$
x_1	1250	$\frac{9}{140}$	$-\frac{1}{28}$	$\frac{11}{24}$	$\frac{1}{28}$ $-\frac{3}{56}$

$$z_1 = z_2 = 0$$

因此, 增加人工变量后, 题目不变

二阶段: $\max z = -(x_1 + x_2 + x_3)$

$$z_1 + 2x_1 + 7.5x_2 + 3x_3 - y_1 = 10000$$

$$z_2 + 20x_1 + 5x_2 + 10x_3 - y_2 = 30000$$

$$x_1, x_2, x_3, y_1, y_2 \geq 0, \quad z_1, z_2 = 0$$

例

	z	x_1	x_2	x_3	y_1	y_2
z	0	-1	-1	-1	0	0
z_1	10000	2	7.5	3	-1	0
z_2	30000	20	5	10	0	-1

由 $\max$ ~~min~~

由阶段一, 去除 z_1, z_2 两列后例

	z'	x_3	y_1	y_2
z'	0	0	0	0
x_2	1000	$\frac{2}{7}$	$-\frac{1}{7}$	$\frac{1}{70}$
x_1	1250	$\frac{11}{24}$	$\frac{1}{28}$	$-\frac{3}{50}$

由于

$$\begin{cases} 1000 = x_2 + \frac{2}{7}x_3 - \frac{1}{7}y_1 + \frac{1}{70}y_2 \\ 1250 = x_1 + \frac{11}{24}x_3 + \frac{1}{28}y_1 - \frac{3}{50}y_2 \end{cases}$$

例

$$\begin{cases} y_1 = \\ y_2 = \end{cases}$$

利用单纯形算法求解, 得

z	2250
x_2	1000
x_1	1250

$x_1 = 1250, x_2 = 1000, x_3 = 0$

最优解: 2250

变化不自变量, 变量变化: 人口数

0	15
0001	5x
0751	1x